Leonard E. van Dyck

lenny.vandyck@gmail.com
levandyck.github.io
Bluesky: @levandyck
Twitter: @levandyck

EDUCATION

PhD in Psychology (Cognitive Computational Neuroscience) — Giessen, Germany
Justus Liebig University Giessen — 09/2023 - present
Advisors: Prof. Katharina Dobs & Prof. Martin Hebart

MSc in Psychology (Cognitive Neuroscience) — Salzburg, Austria
University of Salzburg — 10/2020 - 02/2023
Advisors: Prof. Mario Braun & Prof. Martin Kronbichler

BSc in Psychology — Salzburg, Austria
University of Salzburg — 10/2017 - 07/2020
Advisor: Dr. Walter Gruber

EXPERIENCE

Guest Researcher — Leipzig, Germany
Max Planck Institute for Human Cognitive and Brain Sciences — 10/2023 - 09/2025
Group Leader: Prof. Martin Hebart

Laboratory Manager — Salzburg, Austria
Centre for Cognitive Neuroscience, University of Salzburg — 10/2020 - 02/2023
Method Unit EEG

Teaching Assistant — Salzburg, Austria
Department of Psychology, University of Salzburg — 03/2021 - 07/2022
Bachelor's Seminar "Artificial Intelligence"

Research Assistant — Salzburg, Austria
Laboratory for Sleep, Cognition, and Consciousness Research, University of Salzburg — 03/2020 - 07/2020
Group Leaders: Prof. Manuel Schabus & Prof. Kerstin Hödlmoser

Research Internship — Salzburg, Austria
Department of Psychology, University of Salzburg — 04/2019 - 07/2019
Clinical and Family Therapy, Prof. Katherine Hertlein
(University of Nevada, Las Vegas)

CONTRIBUTIONS

Google Scholar profile (up to date), *equal contribution

Preprints

1. **van Dyck, L. E.** & Dobs, K. (2026). Faces and bodies are increasingly integrated along the visual hierarchy in humans and deep neural networks. *bioRxiv*. https://doi.org/10.64898/2026.02.16.706115

Publications

7. **van Dyck, L. E.**, Hebart, M. N.*, & Dobs, K.* (2026). Multidimensional feature tuning in category-selective areas of human visual cortex. *The Journal of Neuroscience, 46*(31), e0038262026. https://doi.org/10.1523/JNEUROSCI.0038-26.2026

6. **van Dyck, L. E.** & Dobs, K. (2026). Category selectivity as a window into behavioral relevance. *Cognitive Neuroscience, 17*(2), 103-105. https://doi.org/10.1080/17588928.2025.2590656

5. **van Dyck, L. E.**, Bremmer, F., & Dobs, K. (2024). Artificial intelligence meets body sense: task-driven neural networks reveal computational principles of the proprioceptive pathway. *Signal Transduction and Targeted Therapy,* 9(1), 171. https://doi.org/10.1038/s41392-024-01870-9

4. **van Dyck, L. E.** & Gruber, W. R. (2022). Modeling biological face recognition with deep convolutional neural networks. *Journal of Cognitive Neuroscience, 35*(10), 1521-1537. https://doi.org/10.1162/jocn_a_02040

3. **van Dyck, L. E.**, Denzler, S. J., & Gruber, W. R. (2022). Guiding visual attention in deep convolutional neural networks based on human eye movements. *Frontiers in Neuroscience, 16*. https://doi.org/10.3389/fnins.2022.975639

2. **van Dyck, L. E.**, Kwitt, R., Denzler, S. J., & Gruber, W. R. (2021). Comparing object recognition in humans and deep convolutional neural networks – An eye tracking study. *Frontiers in Neuroscience, 15*. https://doi.org/10.3389/fnins.2021.750639

1. Hertlein, K. M. & **van Dyck, L. E.** (2020). Predicting engagement in electronic surveillance in romantic relationships. *Cyberpsychology, Behavior, and Social Networking, 23*(9), 604-610. https://doi.org/10.1089/cyber.2019.0424

Conference Posters

11. **van Dyck, L. E.** & Dobs, K. (2026). Different features explain category selectivity in human visual cortex and deep neural networks. *Cognitive Computational Neuroscience (CCN)*, Aug. 3-6, New York, NY, USA. Short paper

10. **van Dyck, L. E.** & Dobs, K. (2026). Feature-based principles of category selectivity in human visual cortex and deep neural networks. *Vision Sciences Society Meeting (VSS)*, May 15-19, St. Pete Beach, FL, USA.

9. **van Dyck, L. E.** & Dobs, K. (2026). Face and body representations converge along the visual hierarchy in models and cortex. *Workshop on Humans, Deep Networks & Face and Body Recognition*, Mar. 12-13, London, UK.

8. **van Dyck, L. E.** & Dobs, K. (2025). Deep neural networks provide insights into distinct and shared selectivity for faces and bodies in human visual cortex. *Cognitive Computational Neuroscience (CCN)*, Aug. 12-15, Amsterdam, NL. Short paper

7. **van Dyck, L. E.** & Dobs, K. (2025). Deep neural networks provide insights into distinct and shared selectivity for faces and bodies in human visual cortex. *Workshop on CONCEPTS, ACTIONS, and OBJECTS (CAOs), May 7-9, Rovereto, Italy.*

6. **van Dyck, L. E.**, Hebart, M. N.*, & Dobs, K.* (2024). Core neural dimensions of functionally selective areas in the human visual cortex. *Neuro-AI-Talks (NEAT)*, Sept. 02-03, Osnabrück, Germany.

5. **van Dyck, L. E.**, Hebart, M. N.*, & Dobs, K.* (2024). Core neural dimensions of functionally selective areas in the human visual cortex. *European Conference on Visual Perception (ECVP)*, Aug. 25-29, Aberdeen, Scotland.

4. **van Dyck, L. E.**, Hebart, M. N., & Dobs, K. (2024). Core neural dimensions of functionally selective areas in the human visual cortex. *Cognitive Computational Neuroscience (CCN)*, Aug. 6-9, Boston, MA, USA. Short paper

3. **van Dyck, L. E.**, Dobs, K.*, & Hebart, M. N.* (2024). Data-driven voxel decomposition reveals representational dimensions in functionally selective areas. *Workshop on CONCEPTS, ACTIONS, and OBJECTS (CAOs)*, May 9-11, Rovereto, Italy.

2. **van Dyck, L. E.**, Denzler, S. J., & Gruber, W. R. (2022). Analyzing and increasing the similarity of humans and deep convolutional neural networks in object recognition. *European Conference on Visual Perception (ECVP)*, Aug. 28 – Sept. 1, Nijmegen, The Netherlands.

1. **van Dyck, L. E.**, Denzler, S. J., Schöllkopf, C. P., & Gruber, W. R. (2022). Spatial similarities between human eye movements and deep convolutional neural network saliency maps across time. *Salzburg Mind Brain Annual Meeting (SAMBA)*, Jul. 14-15, Salzburg, Austria.

Talks

Invited Talk at Mok Lab (PI: Rob Mok) — Jul. 14, 2026
The Center for Information and Neural Networks (CiNet), Osaka, Japan
Rethinking category selectivity in human visual cortex and deep neural networks

Talk at SFB Workshop "*Categorization in Perception and Action*" — Jun. 30, 2024
Schloss Rauischholzhausen, Germany
Neural representational dimensions capture the nested functional organization of human visual cortex

Invited talk at ViCCo Group (PI: Martin Hebart) — Sept. 29, 2022
Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany
Two approaches to increasing the human-likeness of visual attention in deep neural networks

Theses

van Dyck, L. E. (2023). Unraveling top-down and bottom-up processes in theory of mind with layer fMRI. [Master's Thesis]. https://eplus.uni-salzburg.at/obvusbhs/content/titleinfo/8960559

van Dyck, L. E. & Gruber, W. R. (2020). Seeing Eye-to-Eye? A comparison of object recognition performance in humans and deep convolutional neural networks under image manipulation. *arXiv* [Bachelor's Thesis]. https://doi.org/10.48550/arxiv.2007.06294

Science Communication

van Dyck, L. E. & Dobs, K. (2023). Modellierung der biologischen Gesichtswahrnehmung mit Künstlicher Intelligenz. DER AUGENSPIEGEL (German Journal for Ophthalmologists), Dec. 2023, 50-53. https://www.augenspiegel.com/modellierung-der-biologischen-gesichtswahrnehmung-mit-kuenstlicher-intelligenz/

GRANTS AND AWARDS

Doctoral Scholarship 10/2023 - 09/2026
Studienstiftung des deutschen Volkes e.V.
German Academic Scholarship Foundation

Merit Scholarship 2023
University of Salzburg

ECVP22 Travel Grant 2022
Donders Institute for Brain, Cognition, and Behaviour

ACTIVITIES AND SERVICE

Ad-hoc Reviewer

Journal of Neuroscience, *PLOS Computational Biology*, *Journal of Vision*, *British Journal of Psychology*, *Cognitive Computational Neuroscience Conference*